

# OFM Riaz Rahman Aranya

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Vision and Immersive Realities Lab (VIRLab)  
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## RESEARCH INTERESTS

- Medical Image Segmentation, Vision-Language Models for Clinical Applications, Medical Image Reconstruction, Explainable AI
- Radiology Report Generation, Longitudinal Change Analysis, Semi-Supervised Learning
- Synthetic-to-Real Domain Adaptation, Cross-Modal Image Reconstruction

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## EDUCATION

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| <b>Ph.D. in Computer Science</b><br>The University of Texas at San Antonio, San Antonio, TX<br>Expected Graduation: 2028   GPA: 4.00/4.00   Advisor: <a href="#">Dr. Kevin Desai</a> | <b>Aug 2023 – Present</b>  |
| <b>B.Sc. in Computer Science and Engineering</b><br>Khulna University of Engineering & Technology, Bangladesh<br>GPA: 3.28/4.00                                                      | <b>Mar 2016 – Mar 2020</b> |

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## PUBLICATIONS

### Journal Papers

1. A. Bagherian, **O.F.M. R. R. Aranya**, A. Kosub, et al. “Comprehensive Benchmarking of Deep Learning Approaches for Automated Astrocyte Segmentation in Traumatic Brain Injury.” *Journal of Neuropathology and Experimental Neurology*, 2025. [DOI](#)
2. P. Saha, M. S. Sadi, **O.F.M. R. R. Aranya**, S. Jahan, F. Islam. “COV-VGX: An Automated COVID-19 Detection System Using X-ray Images and Transfer Learning.” *Informat-ics in Medicine Unlocked*, 26:100741, 2021. [DOI](#)

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### Conference Papers

3. **O.F.M. R. R. Aranya**, P. Rad, K. Desai. “GRCD: Grounded Region Change Detection for Multi-Finding Chest X-Ray Pairs.” *British Machine Vision Conference (BMVC) 2026* (Submitted).
4. M. B. Akram, A. H. M. N. Sakib, **O.F.M. R. R. Aranya**, R. Wijewickrama, K. Desai, M. Jadliwala. “ThermalTap: Passive Application Fingerprinting in VR Headsets via Thermal Side Channels.” *Network and Distributed System Security (NDSS) Symposium 2027* (Submitted). [\[arXiv\]](#)
5. **O.F.M. R. R. Aranya**, K. Desai. “SRA-Seg: Synthetic to Real Alignment for Semi-Supervised Medical Image Segmentation.” *International Conference on Pattern Recognition (ICPR)*, 2026. (To appear) [\[arXiv\]](#) [\[Project\]](#) [\[Code\]](#)
6. **O.F.M. R. R. Aranya**, K. Desai. “TRACE: Temporal Radiology with Anatomical Change Explanation for Grounded X-ray Report Generation.” *International Conference on Pattern Recognition (ICPR)*, 2026. (To appear) [\[arXiv\]](#) [\[Project\]](#) [\[Code\]](#)

7. **O.F.M. R. R. Aranya**, K. Desai. “To Agree or To Be Right? The Grounding-Sycophancy Tradeoff in Medical Vision-Language Models.” *CVPR Workshop (Med-Reasoner)*, 2026. [\[Oral\]](#) [\[Best paper award\]](#) [\[PDF\]](#) [\[arXiv\]](#) [\[Project\]](#) [\[Code\]](#)

8. A. N. M. Sakib, **O.F.M. R. R. Aranya**, K. Desai, Z. Zhang. “Toward Faithful Segmentation Attribution via Benchmarking and Dual-Evidence Fusion.” *CVPR Workshop (HOW)*, 2026. [\[PDF\]](#) [\[arXiv\]](#) [\[Code\]](#)

9. S. Das, **O.F.M. R. R. Aranya**, N. N. Labiba. “Brain Tumor Classification Using Convolutional Neural Network.” *IEEE ICASERT*, 2019. [DOI](#)

**RESEARCH  
EXPERIENCE**

**Graduate Research Assistant** | *VIRLab, UTSA* **Jan 2026 – Present**  
Advisor: [Dr. Kevin Desai](#)

- Research on medical image segmentation, cross-modal reconstruction, and vision-language models for clinical applications.
- Key results: SRA-Seg (89.34% Dice, ACDC, 10% labels), TRACE (90.2% grounding accuracy), TRACE-R (substantially outperforming TRACE across all clinical metrics).
- Discovered the grounding-sycophancy tradeoff in medical VLMs ( $CSI < 0.35$  across all models).
- Clinical collaborations: OCT-to-histology cardiac reconstruction with a practicing cardiologist; cross-institutional neuropathology segmentation.

**TEACHING  
EXPERIENCE**

**Instructor** **Aug 2023 – Dec 2025**  
*Department of Computer Science, The University of Texas at San Antonio*

- CS 2113 – Fundamentals of Object-Oriented Programming (Fall 2025)
- CS 3443 – Application Programming (Fall 2024)

**Teaching Assistant**  
*Department of Computer Science, The University of Texas at San Antonio*

- CS 4843 – Cloud Computing (Summer 2025, Spring 2025)
- CS 3853 – Computer Architecture (Summer 2024)
- CS 3443 – Application Programming (Spring 2024)
- CS 2233 – Discrete Math Structures (Fall 2023)

**INDUSTRY  
EXPERIENCE**

**Software Engineer** | *RedDot Digital Ltd, Dhaka, Bangladesh* **Mar 2023 – Jul 2023**  
Designed solution architecture and developed RESTful APIs for enterprise applications.

**Jr. Software Engineer** | *Nascenia Ltd, Dhaka, Bangladesh* **Dec 2020 – Feb 2023**  
Full-stack developer (Ruby on Rails). Designed solution architectures, delivered multiple production systems, and mentored junior engineers.

## ACADEMIC SERVICE

### Peer Review

- Med-Reasoner Workshop, CVPR 2026 (2 papers reviewed)
  - IEEE Signal Processing Letters (7 paper reviewed)
  - Informatics in Medicine Unlocked, Elsevier (1 paper reviewed)
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## LEADERSHIP & SERVICE

- Bangladesh Student Association at UTSA – **Treasurer** (2024–2025). Managed finances and event budgeting for 100+ member community.
  - Jashore Association of KUET – **Vice President** (2019–2020), **General Secretary** (2018–2019).
  - HACK (Hardware Acceleration Club of KUET) – **General Secretary** (2019–2020).
  - Dream (Voluntary Blood Donation Society of KUET) – Asst. General Secretary (2019–2020).
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## HONORS & AWARDS

- **Best Paper Award**, Med-Reasoner Workshop, CVPR 2026.
  - Published work (COV-VGX) recognized under the **United Nations Sustainable Development Goal 3** (Good Health and Well-being).
  - 1st Runner Up, LICT Project Showcasing, Jashore IT Park, 2019.
  - Board Scholarship (General Merit), Jashore Board, 2015.
  - Champion, High Jump, KUET (2019, 2020); 2nd Runner Up, Long Jump (2019).
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## PROFESSIONAL MEMBERSHIP

- IEEE Member (ID: 99988407), June 2024 – Present.
  - CVF Member (ID: b7b14c38cc), March 2024 – Present.
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## TECHNICAL SKILLS

**Languages:** Python, Ruby, Java, C, C++  
**ML/DL:** PyTorch, TensorFlow, HuggingFace Transformers  
**Web:** Ruby on Rails, JavaScript, HTML/CSS  
**Tools:** Docker, Git, Linux, LaTeX, MySQL